# PROJECT REPORT - CUSTOM WC & CUSTOM RM

Prabhujit Singh
Adm No: 23JE0723
Group 25

## Introduction

This project focuses on developing simplified versions of essential UNIX utilities, namely wc (word count) and rm (remove), using the C programming language. The primary goal is to understand how low-level system tools are designed and implemented internally. These utilities are integrated with a custom-built shell, allowing execution in a way similar to standard Linux commands. Through this project, key concepts such as file handling, command-line argument parsing, and system calls are explored in depth. It also provides practical exposure to building efficient command-line tools, which are fundamental in operating systems and real-world software development.

## Features of custom_wc

1. Counts number of lines in a file using -l flag
2. Counts number of words using -w flag
3. Counts number of characters using -c flag
4. Supports multiple file inputs in a single command
5. Handles standard input (stdin) when no file is provided

## Features of custom_rm

1. Deletes single or multiple files efficiently
2. Interactive mode (-i) asks confirmation before deletion
3. Displays proper error messages for invalid files
4. Prevents accidental deletion through confirmation prompts
5. Supports batch deletion of multiple files

## Concepts Used

1. File handling using fopen(), fgetc(), fclose()
2. Command-line argument parsing using argc and argv

3 System calls like remove() for file deletion
4 Error handling using perror() and validation checks

## Screenshots

The following screenshots demonstrate execution of the implemented utilities under different scenarios:

```
custom_shell:~/G25_OS-Project/G25_Project2_1$ ./custom_rm test.t
xt
Deleted: test.txt
```

*Test 1: File deletion using custom_rm*

```
custom_shell:~/G25_OS-Project/G25_Project2_1$ touch test.txt
custom_shell:~/G25_OS-Project/G25_Project2_1$ ls
Adyant.pdf  custom_cat.c   custom_rm.c     custom_wc
Makefile    custom_grep.c  custom_shell    custom_wc.c
README.md   custom_rm      custom_shell.c  test.txt
custom_shell:~/G25_OS-Project/G25_Project2_1$ ./custom_rm test.t
xt
File deleted successfully
custom_shell:~/G25_OS-Project/G25_Project2_1$ ls
Adyant.pdf  custom_cat.c   custom_rm.c     custom_wc
Makefile    custom_grep.c  custom_shell    custom_wc.c
README.md   custom_rm      custom_shell.c
custom_shell:~/G25_OS-Project/G25_Project2_1$
```

*Test 2: Basic word count output*

```
custom_shell:~/G25_OS-Project/G25_Project2_1$ ./custom_wc README
.md
Lines: 253 Words: 1195 Characters: 9180
```

*Test 3: Advanced count output*

```
custom_shell:~/G25_OS-Project/G25_Project2_1$ touch test.txt
custom_shell:~/G25_OS-Project/G25_Project2_1$ ./custom_wc
hello world
1 2 12
```

*Test 4: Standard input handling*

```
custom_shell:~/G25_OS-Project/G25_Project2_1$ touch a.txt b.txt
custom_shell:~/G25_OS-Project/G25_Project2_1$ ./custom_rm a.txt
b.txt
Deleted: a.txt
Deleted: b.txt
```

*Test 5: Multiple file deletion*

## Conclusion

This project successfully demonstrates the implementation of UNIX-like utilities and provides a strong foundation in system-level programming. It enhances understanding of file operations, command-line interfaces, and error handling techniques. The integration with a custom shell further strengthens knowledge of process execution and command workflows. Overall, this project serves as an important step toward mastering operating systems concepts and developing efficient, real-world software tools.